

Essential Summary

Predicting Homoclinic and Heteroclinic Bifurcation of Generalized Duffing–Harmonic–van de Pol Oscillator

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Qualitative Theory of Dynamical Systems, Vol. 15, No. 1, 19–37 (2016) DOI: 10.1007/s12346-015-0138-z

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Please cite the published version.

1 Background and Motivation

The Duffing–harmonic oscillator, characterized by a rational restoring force,

$$\ddot{x} + \frac{\lambda x + \delta x^3}{1 + \nu x^2} = 0 \quad (1)$$

bridges two well-studied regimes: for small x , it behaves as a Duffing oscillator ($\ddot{x} + \delta x^3 \approx 0$); for large x , it approximates a linear harmonic oscillator ($\ddot{x} + (\delta/\nu)x \approx 0$). This dual nature makes it an excellent model for systems with amplitude-dependent stiffness transitions.

While the conservative Duffing–harmonic oscillator had been extensively studied via harmonic balance, homotopy analysis, and energy balance methods, the **damped** version—the generalized Duffing–harmonic–van de Pol oscillator—

$$\ddot{x} + \frac{\lambda x + \delta x^3}{1 + \nu x^2} = \varepsilon(\mu + \mu_1 x + \mu_2 x^2 + \mu_4 x^4)\dot{x} \quad (2)$$

had received almost no attention. This paper fills that gap by applying the quadratic generalized harmonic function perturbation method to predict both homoclinic and heteroclinic bifurcations in this oscillator.

2 Quadratic Generalized Harmonic Function

2.1 Solution Construction

Building on the original GHFP method (JSV 2013), this paper constructs solutions in the form:

$$x(t) = a \sin^2 \phi(t) + b \quad (\text{homoclinic orbits}) \quad (3)$$

$$x(t) = a \cos^2 \phi(t) + b \quad (\text{heteroclinic orbits}) \quad (4)$$

The nonlinear time transformation:

$$\frac{d\phi}{dt} = \Phi(\phi), \quad \Phi(\phi + \pi) = \Phi(\phi) \quad (5)$$

maps the time domain to an angular domain where the solutions have a symmetric form.

2.2 Homoclinic Orbits

For a saddle point $H(h, 0)$, the initial condition c satisfies:

$$\int_h^c \frac{\lambda x + \delta x^3}{1 + \nu x^2} dx = 0 \quad (6)$$

and the homoclinic solution must satisfy $x(\phi_{\pm\infty}) = h$. The angular velocity is obtained as:

$$\Phi(\phi) = \sqrt{\frac{2[V(b) - V(a \sin^2 \phi + b)]}{(x'(\phi))^2}} \quad (7)$$

with $\Phi(0) = \Phi(\pi) = \sqrt{|g(h)/2a|} = 0$ ensuring infinite period at the saddle point.

2.3 Heteroclinic Orbits

For two distinct saddle points $H_1(h_1, 0)$ and $H_2(h_2, 0)$:

$$\lim_{t \rightarrow -\infty} x(t) = h_1, \quad \lim_{t \rightarrow +\infty} x(t) = h_2 \quad (8)$$

The constants a and b are determined by $\phi = 0 \rightarrow x = h_1$ and $\phi = \pi/2 \rightarrow x = h_2$. The angular velocity satisfies $\Phi(0) = \Phi(\pi/2) = 0$.

3 Perturbation Procedure

3.1 Expansion and Key Assumption

When $\varepsilon \neq 0$:

$$a = a_0 + \varepsilon a_1 + \dots, \quad \Phi = \Phi_0 + \varepsilon \Phi_1 + \dots \quad (9)$$

The determining insight: **the n th-order approximate solution maintains the same functional form as the generating solution:**

$$x_n = a_n \sin^2 \phi \quad (\text{homoclinic}) \quad (10)$$

$$x_n = a_n \cos^2 \phi \quad (\text{heteroclinic}) \quad (11)$$

This eliminates the cumbersome integrals that plague classical perturbation methods.

3.2 Homoclinic Bifurcation Prediction

The first-order consistency condition yields:

$$\mu_c = - \frac{\int_0^\pi \Phi_0(\mu_1 x_0 + \mu_2 x_0^2 + \mu_4 x_0^4)(\Phi_0 x'_0) x'_0 d\varphi}{\int_0^\pi \Phi_0^2(x'_0)^2 d\varphi} \quad (12)$$

This formula directly predicts the critical parameter value μ_c at which homoclinic bifurcation occurs.

4 Key Results

4.1 Example 1: Duffing–Harmonic Oscillator with (μ_1, μ_2, μ_4) Damping

For the system with $\lambda = \delta = \nu = 1$, the GHFP method accurately predicts homoclinic orbits across a wide range of ε values.

4.2 Example 2: Multiple Coexisting Homoclinic Orbits

For a system with different parameters, the method captures two types of homoclinic orbits—one enclosing a single well and another enclosing both wells—at the same parameter values.

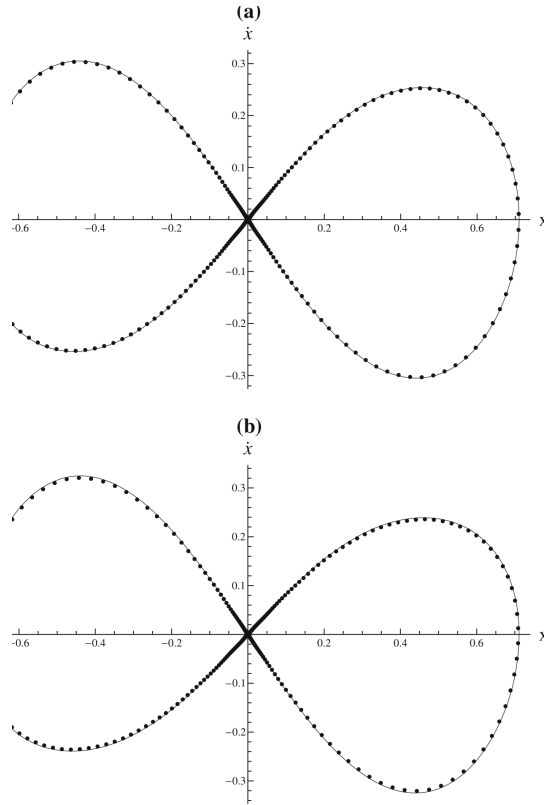


Figure 1: **Figure 1.** (Original Fig. 1) Homoclinic orbits for $\varepsilon = 1.2, 2.0$. Solid lines: Runge–Kutta method. Dotted lines: GHFP method. Excellent agreement demonstrates the effectiveness of the quadratic generalized harmonic function approach for rational restoring forces.

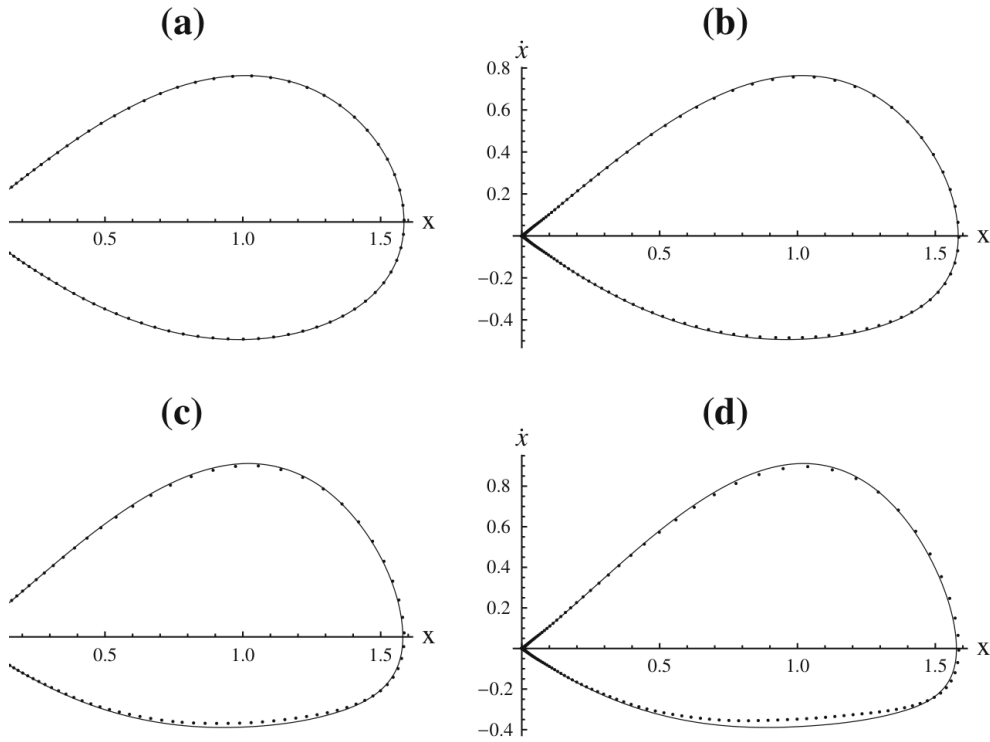


Figure 2: **Figure 2.** (Original Fig. 2) Homoclinic orbits for $\varepsilon = 0.5, 1.0, 1.5, 2.0$. The method successfully tracks the orbits across a wide range, maintaining accuracy even at $\varepsilon = 2.0$.

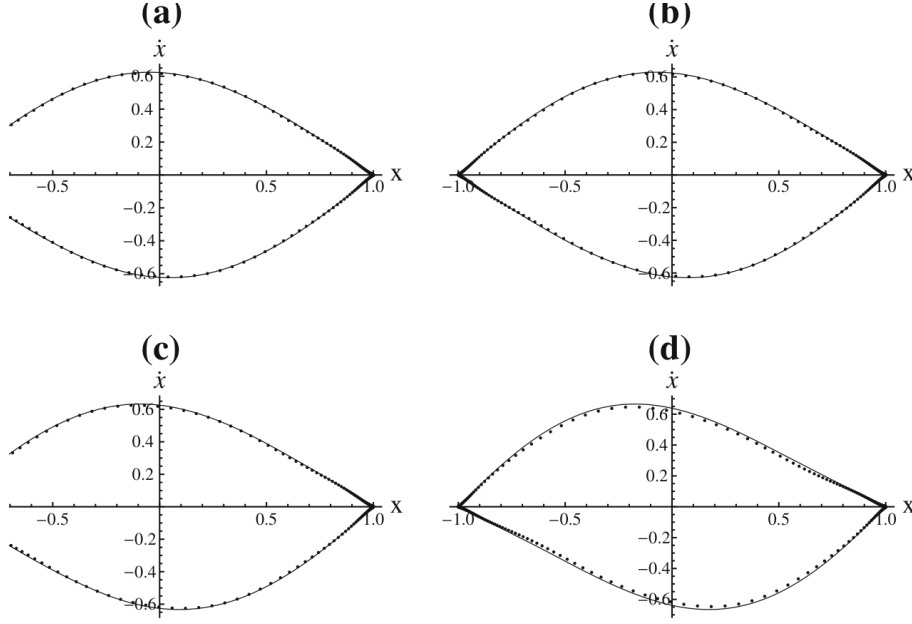


Figure 3: **Figure 3.** (Original Fig. 4) Heteroclinic orbits connecting two distinct saddle points for $\varepsilon = 0.5, 0.8, 1.0, 2.0$. The method captures cross-saddle connections with remarkable accuracy.

4.3 Example 3: Heteroclinic Bifurcation Prediction

For the heteroclinic case, the method predicts critical parameter values:

$$\mu_c = -0.1979871606 \quad (\varepsilon = 0.5), \quad \mu_c^{(\text{numerical})} = -0.1977132821 \quad (13)$$

The relative error is approximately 0.14%.

5 Significance

1. **First application of GHFP to rational restoring forces:** Unlike the polynomial Helmholtz–Duffing oscillator (JSV 2013), the rational form $\frac{\lambda x + \delta x^3}{1 + \nu x^2}$ introduces fundamentally different Fourier coefficient computations. The method’s success here demonstrates its versatility.
2. **Simultaneous homoclinic and heteroclinic prediction:** The same framework handles both types of global bifurcations without modification—a capability not available with classical perturbation methods.
3. **Connection to the MGHP series:** This paper bridges the original GHFP method (2013) and the modern MGHP series (2024–2026), showing the evolution from polynomial to rational restoring forces.

6 Limitations

- First-order perturbation limits accuracy when $\varepsilon > 2$ for some parameter regions.
- The method requires analytical knowledge of the saddle points, which may become algebraically intractable for more complex restoring forces.
- Only single-degree-of-freedom systems are considered.

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